PromoFinder

Problem 1 · Find the Promos · solution note

Tested on all 50 episodes of "I Married a Broken Soldier. He Was a Duke"

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PromoFinder reads the screenplays of a whole microdrama and returns a ranked slate of 20–60 s promo clips. Each clip comes with its episode, the start and end of every segment, the hook line, the cliffhanger line and a one-line reason, and every timestamp and quote is checked against the screenplay.

50 → 387 → 12
episodes, candidate clips, recommended slate

8/8 · 7/7
human-picked clips found (development sets)

0 / 387
shown quotes outside their clip (26/376 before the strict validator)

70 / 70
automated checks passed: core, formats, adapter, browser

Post these first (top of the 12-clip slate: each opens on a hook inside 3 seconds and stops on an unresolved moment)

#1 Ep36 02:26–03:16 · 50s
"What wedding?"
CUTS ON ...their dead Duke is coming home.
Why: Brilliant emotional and narrative arc from learning about the wedding and execution to...

#2 Ep43 00:00–00:21 · 21s
"You're lying!"
CUTS ON No...
Why: Great escalation from the guards refusing arrest to the reading of the treasonous...

#3 Ep23 01:19–02:11 · 52s
"Someone is poisoning her."
CUTS ON Loud, metallic clack of the door latch turning, followed by the slow...
Why: The ultimate master promo: opens with the jaw-dropping arsenic revelation and builds...

The problem (Kuku, Problem 1)
An editor watches every episode to pick promo moments by hand, and new shows arrive weekly. A promo is a 20–60 s clip that opens on a hook in the first 3 seconds, ends on an unresolved cliffhanger and makes sense to someone who has never seen the show. The task: find them, give episode and start/end of every segment, rank them, and check where the tool picks badly. Text only: no editing, music or overlays. **Why I chose it:** its output can be checked by code (every time and quote is verifiable), it repeats with every new show, and the full pack lets me test it for real.

What PromoFinder does
Reads every episode, builds candidate windows, has a model judge them, makes every timestamp and quote verifiable, ranks the best clips of the whole show head to head, and hands over a slate of 12 plus all 387 ranked candidates. A dashboard lets an editor preview, shortlist, export and run it on a new show. The slate holds at most two clips per episode (12 episodes appear).

01 Architecture and workflow

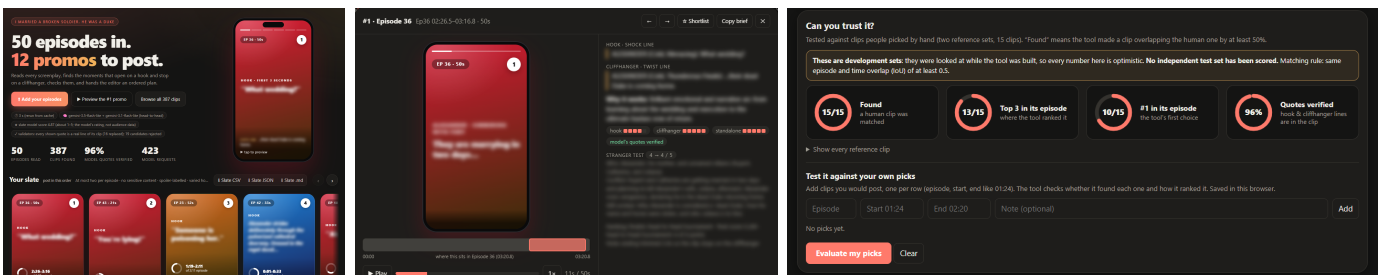
1 Parse → 2 Funnel → 3 Judge → 4 Stranger test → 5 Validate → 6 Rank → 7 Slate → 8 Editor

plain code · AI model · code + AI · human

1·2	code	Parse any supported format; score every 20–60 s window on beat boundaries; keep 12 per episode.
3	AI, 1 call per episode	Reads the episode, quotes hook and cliffhanger, rates them 1–5, writes the reason, may add windows the rules missed.
4	AI, finalists only	Stranger test: does someone who has seen nothing follow the clip? Tries a setup segment if not.
5	code (strict)	Snaps to beats, enforces 20–60 s, dedupes, makes every quoted line a real line of its clip.
6·7	AI + code	The 60 best clips of the whole show (at most 2 per episode) meet head to head; code picks a varied slate and trims it to open on the hook.
8	human	An editor accepts, reorders or drops in the dashboard; every clip has a clip_id for tracking.

Technical decisions that matter

- **Code decides, the model proposes.** Timestamps, quotes, length, duplicates and sensitive-content holds are checked by code; a model line that fails is pulled in, replaced or rejected, never trusted.
- **Head to head, not absolute scores.** Model 1–5 ratings bunch up across episodes, so the finalists are compared in pairs, each pair in both orders.
- **Not tied to one show.** Generic word lists, five input formats, swappable model, and a quality report that needs no human labels.



Real screenshots of the working app: dashboard and slate, the video-style preview, and the evaluation view (script text blurred).

02 How it was evaluated

Setup. The tool was run on all 50 episodes of the pack. Two reference sets exist: set A, 8 clips I picked by hand while the tool was being built, and set B, 7 clips drafted by the AI assistant from the scripts. **Both are development sets:** I looked at them while designing the tool, so every number below is optimistic, and **no independent test set has been scored**. Both files are frozen (SHA-256) and labelled so; a fresh, boundary-checked set can be scored with one command. **Matching rule:** same episode and time-span IoU of at least 0.5, because human and tool boundaries rarely coincide (only 4 of A's and 1 of B's picks sit on exact screenplay beat boundaries). Sets are tiny: one clip moves a recall figure by 12–14 points.

Measure	Result (measured on this run)
Human clips found among all clips	A: **8/8** · B: **7/7**
Inside the overall top 10 / 12 / 20 (A)	12% / 12% / 50%
Inside the overall top 10 / 12 / 20 (B)	29% / 43% / 43%
Best clip of its own episode: #1 / top 3	A: 50% / 88% · B: 86% / 86%
MRR / median rank of the best matching clip	A: 0.073 / 25.0 · B: 0.088 / 31
Slate clips that match a human pick (precision, a lower bound)	A: 1/12 · B: 3/12
Shown hook/cliffhanger line outside its clip	**0/387** (model's 774 quoted lines: 666 right, 92 pulled in, 16 replaced)
Hook within 3 s / clip ends on its cliffhanger	all clips: 99% / 82% · slate: 100% / 100%
Head-to-head judge: pairs split between the two orders	21% (a second model 20%; the two models agree, Spearman 0.76)

Failure analysis

- **Cross-episode ranking is unreliable.** With this show's own words in the funnel 3 of 8 human picks reached the top 10; after I made the funnel generic it was 1 of 8. I treat that swing as noise and claim no improvement.
- **Inside an episode, set A:** the tool's top clip is my pick in 4 of 8 episodes. In Ep 3 and Ep 18 its top clip is a different scene (0–7% overlap with mine); in Ep 41 and Ep 42 it is a neighbouring cut (38%). My picks there are its 2nd or 3rd clip.
- **Ep 3 (the stranger's proposal)** is found but ranks 291 of 387: the model rates its cliffhanger 1/5 because the scene resolves.
- The spoiler check cannot see that the show's title gives the twist away; the model's "why" texts praise almost everything.

Limitations and assumptions

- Evaluation is on development sets only; model scores are model judgements, not audience behaviour.
- Of the ten dimensions the brief suggests, the model rates hook, cliffhanger and standalone clarity; a newcomer test, a spoiler label and duration add three more; emotional intensity, curiosity, retention and social fit are not scored separately.
- Text only; no video, audio or viewer data. Word lists are English. Clips with estimated times (scripts without timestamps) are flagged as approximate.
- Assumptions where the brief was silent: a promo is one stretch of one episode (joining a setup from the previous episode was tried and did not help); the slate size is 12; screenplay timestamps are exact; a viewer who clicks a promo starts at Episode 1 (a hypothesis).

03 Scaling, tools, and business impact

Scaling to many dramas

- 423 model requests for a 50-episode show (about 30 minutes at the free tier's 15 calls a minute); cached steps are free, so a rerun costs nothing. At 250 shows a month that is about 110,000 requests: use a paid or batch endpoint.
- No source change per show: folder or ZIP upload, five input formats, a model and endpoint set in one config file, per-stage rate limits, and a quality report for shows nobody has labelled.
- Stress-tested on a disguised copy (new names, odd file names, missing headers, no timestamps, inline times, .srt).

Tools and one AI mistake

Claude Code (planning, code, tests), Gemini 3.5 Flash-Lite and 3.1 Flash-Lite (runtime, free tier), Playwright (browser tests), ReportLab (this PDF). **Mistake:** the AI and I reported "top-10 recall 1/8 → 3/8" as an improvement; removing this show's own words from the funnel showed it was noise and partly fitted to one show.

Does it help? A real experiment

Post the slate (arm A) and the editor's usual picks or random clips (arm B) on the same channels in the same weeks; tag each link with the clip's **clip_id** (in every CSV export).

3-second hold, completion	A ≥ B (Kuku's benchmark 50–60%)
Installs per 1,000 views	main metric: A beats B by a margin set in advance
Ep 1 started; Ep 1 → 3 retention	guardrail: no clickbait
Editor keeps / drops	tells us which rule to fix

Proposed decision rules, not results: I have no Kuku audience data.

Prototype access. Repository: github.com/rajeshroj5628/rajesh-kuku-assigment. **python app.py --scripts <folder>** opens the dashboard at http://127.0.0.1:8000 (upload files or a ZIP, Run, preview, export); **python run_pipeline.py --input <folder>** runs the same pipeline; **python eval/run_eval.py** and **python eval/run_all_tests.py** reproduce the evaluation and tests. Raw outputs of the full 50-episode run: the out/ folder, shared by a restricted link. The input scripts are confidential and are not in the repository.